

Basic Counting

COMP2711 Discrete Mathematical Tools

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125

王牌冰店

Golden Sun Mel Restaurant



20週年紀念
20週年紀念
20週年紀念

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How many different combos are there?

Choose at least 1 entree, up to 2

- BBQ pork
- fried eggplant and pepper
- plain chicken
- soy sauce chicken
- roast duck

Choose 1 vegetable

- green beans
- cabbage
- broccoli

Choose 1 drink

- Coke
- lemon tea
- milk tea
- coffee
- water

Method 1: List every combo

1. BBQ pork
green beans, Coke
2. fried eggplant and pepper
cabbage, lemon tea
3. plain chicken
broccoli, milk tea
4. soy sauce chicken
green beans, coffee
5. roast duck
cabbage, water
6. BBQ pork + fried eggplant and pepper
broccoli, lemon tea
7. BBQ pork + roast duck
green beans, milk tea
8. fried eggplant and pepper + plain chicken
cabbage, coffee
9. plain chicken + soy sauce chicken
broccoli, Coke
10. soy sauce chicken + roast duck
green beans, water

Method 1: List every combo

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green beans, Coke
2. fried eggplant and pepper
cabbage, lemon tea
3. plain chicken
broccoli, milk tea
4. soy sauce chicken
green beans, coffee
5. roast duck
cabbage, water
6. BBQ pork + fried eggplant and pepper
broccoli, lemon tea
7. BBQ pork + roast duck
green beans, milk tea
8. fried eggplant and pepper + plain chicken
cabbage, coffee
9. plain chicken + soy sauce chicken
broccoli, Coke
10. soy sauce chicken + roast duck
green beans, water

And many more ...

The full list is too long for one menu page.

Method 2: Count the choices

(5 single entrees + 10 entree pairs) $\times$ 3 vegetables $\times$ 5 drinks

$$15 \cdot 3 \cdot 5 = \boxed{225 \text{ combo meals}}.$$

One particular combo



A customer eats one combo here every day for 1,000 consecutive days.

Consider **soy sauce chicken + cabbage + milk tea**.

How many times must he order it?

One particular combo



A customer eats one combo here every day for 1,000 consecutive days.

Consider **soy sauce chicken + cabbage + milk tea**.

How many times must he order it?

Possibly 0 times.

He may hate cabbage and never order it.

The most frequent combo



Now we learn that **soy sauce chicken + cabbage + milk tea** was his favorite and most frequently ordered combo.

At least how many times did he order it?

Can we inspect every possibility?

There are 225 choices each day:

225^{1000} ordering sequences.

No. The number is astronomical.

The most frequent combo



Now we learn that **soy sauce chicken + cabbage + milk tea** was his favorite and most frequently ordered combo.

At least how many times did he order it?

Generalized Pigeonhole Principle

1,000 meals are assigned to 225 combos:

$$\left\lceil \frac{1000}{225} \right\rceil = 5$$

He ordered it at least 5 times.

SECTION 1

Modeling the problem

Two kinds of questions

Exact count

How many valid outcomes are possible?

Two kinds of questions

Exact count

How many valid outcomes are possible?

Guaranteed repetition

What must eventually repeat?

SECTION 2

Exact counts

Product rule

MUST UNDERSTAND Multiply when one outcome is built through successive choices.

$$\boxed{n_1} \times \boxed{n_2} \times \boxed{\dots} \times \boxed{n_k}$$

$$\boxed{n_1 n_2 \dots n_k}$$

Example 1: Student IDs

A campus ID has two uppercase letters, followed by three digits.

AB123 HK905 CS271

Example 1: Student IDs

A campus ID has two uppercase letters, followed by three digits.

AB123 HK905 CS271

The first digit cannot be 0. How many IDs are possible?

Example 1: Student IDs



Example 1: Student IDs



$$26 \cdot 26 \cdot 9 \cdot 10 \cdot 10 = 608,400.$$

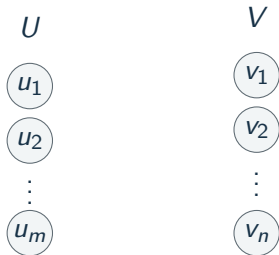
Example 2: Counting functions

Let $U = \{a, b\}$ and $V = \{1, 2, 3\}$.

How many functions $f : U \rightarrow V$ are possible?

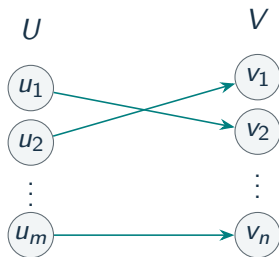
Example 2: Counting functions

Let $|U| = m$ and $|V| = n$.



Example 2: Counting functions

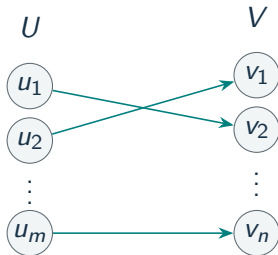
Let $|U| = m$ and $|V| = n$.



Each of the m inputs chooses one of n outputs.

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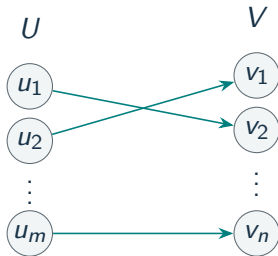
n^m functions $U \rightarrow V$.

For these sets: $3^2 = 9$.

Example 3: Counting injective functions

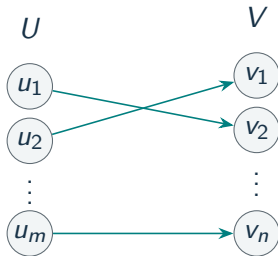
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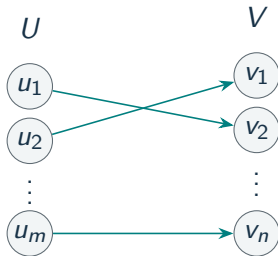


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No two inputs may choose the same output.

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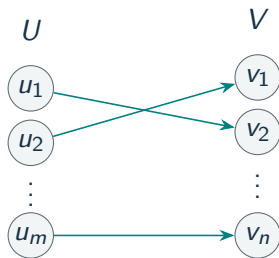
The choices decrease:

$$n(n-1) \cdots (n-m+1).$$

For these sets: $3 \cdot 2 = \boxed{6}$.

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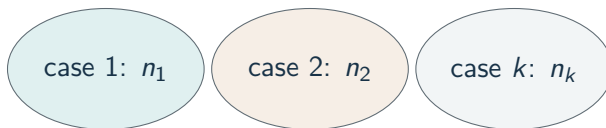
$$n(n-1) \cdots (n-m+1).$$

For these sets: $3 \cdot 2 = \boxed{6}$.

If $m > n$, an injective function is impossible.

Sum rule

MUST UNDERSTAND Add when the valid outcomes are split into disjoint cases.



$$n_1 + n_2 + \cdots + n_k$$

Example 2: Getting to campus

Choose either a bus or a ferry.

Bus: 3 routes, each with 4 departure times.

Ferry: 2 routes, each with 3 departure times.

How many travel plans are possible?

Example 2: Getting to campus

Bus

3 routes · 4 times = 12

Ferry

2 routes · 3 times = 6

Example 2: Getting to campus

Bus

3 routes · 4 times = 12

Ferry

2 routes · 3 times = 6

$$12 + 6 = \boxed{18}.$$

How many different combos are there?

Choose at least 1 entree, up to 2 different entrees.

Case 1: Choose 1 entree

5 choices

Case 2: Choose 2 entrees

count this next

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$5 + (\text{number of two-entree pairs}).$

Two different entrees

Product rule: choose the first entree, then the second.

$$5 \cdot 4 = 20 \text{ ordered selections.}$$

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eggplant + duck and **duck + eggplant**

describe the same pair.

Two different entrees

Product rule: choose the first entree, then the second.

$$5 \cdot 4 = 20 \text{ ordered selections.}$$

eggplant + duck and **duck + eggplant**

describe the same pair.

Every pair was counted twice, so

$$\frac{5 \cdot 4}{2} = \boxed{10 \text{ two-entree choices}}.$$

All combo meals

Entrees

$$5 + 10 = 15$$

Vegetable

3 choices

Drink

5 choices

All combo meals

Entrees

$$5 + 10 = 15$$

Vegetable

3 choices

Drink

5 choices

$$15 \cdot 3 \cdot 5 = \boxed{225 \text{ combo meals}}.$$

SECTION 3

Reorganizing exact counts

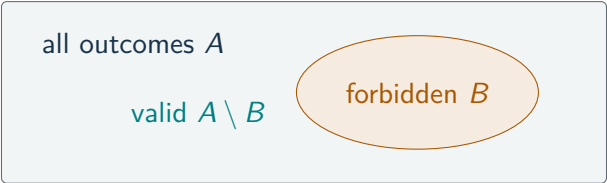
Complement counting

MUST UNDERSTAND Count all easy outcomes, then subtract the forbidden outcomes.

all outcomes A

valid $A \setminus B$

forbidden B



$$|A \setminus B| = |A| - |B|.$$

“At least one” often has the simple complement “none.”

Example 3: Password policy

A password has six characters, each an uppercase letter or a digit.

How many contain at least one digit?

Example 3: Password policy

The complement is: **all six characters are letters.**

Example 3: Password policy

All alphanumeric passwords	36^6
Forbidden all-letter passwords	26^6

$$36^6 - 26^6 = \boxed{1,867,866,560}.$$

Example 3: Password policy

All alphanumeric passwords	36^6
Forbidden all-letter passwords	26^6

$$36^6 - 26^6 = \boxed{1,867,866,560}.$$

Do not choose one digit position first: passwords with several digits are counted several times.

Cafe example: Duck or coffee

Entrees

Choose 1 or 2 different
entrees from 5.

Vegetable

Choose 1 from 3.

Drink

Choose 1 from 5.

The cafe offers **225 different combo meals**.

A customer will not order any combo containing **roast duck** or **coffee**.

How many different combos may he order?

Counting the forbidden combos

Duck appears in 5 entree choices: alone, or paired with one of 4 other entrees.

Forbidden combos	Count
contain duck	$5 \cdot 3 \cdot 5 = 75$
contain coffee	$15 \cdot 3 \cdot 1 = 45$
contain both duck and coffee	$5 \cdot 3 \cdot 1 = 15$

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The 15 combos containing both were subtracted twice, so **add them back once**.

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The 15 combos containing both were subtracted twice, so **add them back once**.

$$225 - 75 - 45 + 15 = \boxed{120 \text{ allowed combos}}.$$

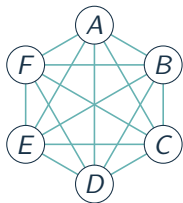
This overlap correction is the **inclusion–exclusion principle**.

Division rule

MUST UNDERSTAND Divide only when every object has the same number of descriptions.

Six people attend a meeting. Every pair of distinct people shakes hands exactly once.

How many handshakes occur?



K_6 : one edge per handshake.

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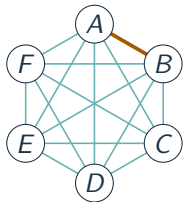
Six people attend a meeting. Every pair of distinct people shakes hands exactly once.

How many handshakes occur?

Choose the first person, then the second:

$$6 \cdot 5 = 30 \text{ ordered descriptions.}$$

(A, B) and (B, A) describe the **same edge**.
Each handshake is counted twice.



K_6 : one edge per handshake.

Division rule

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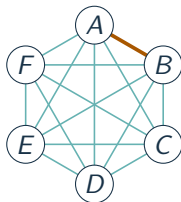
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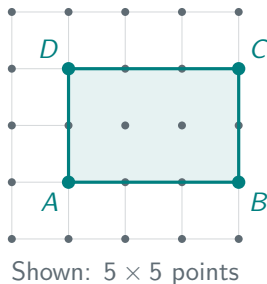


K_6 : one edge per handshake.

$$\frac{6 \cdot 5}{2} = \boxed{15 \text{ handshakes}}.$$

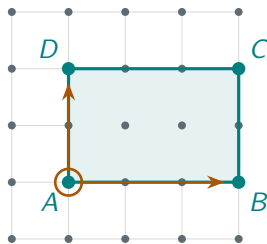
Division rule: Rectangles on a grid

On an $n \times n$ grid of **points** ($n \geq 2$), how many rectangles have grid-point vertices and only horizontal or vertical sides?



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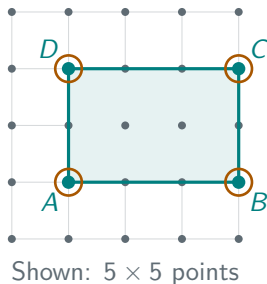
Shown: 5×5 points

Count with a chosen starting vertex

1. Choose the starting point: n^2 choices.
2. Choose a different column: $n - 1$ choices.
3. Choose a different row: $n - 1$ choices.

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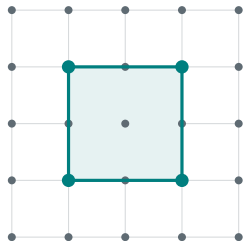
Count with a chosen starting vertex

1. Choose the starting point: n^2 choices.
2. Choose a different column: $n - 1$ choices.
3. Choose a different row: $n - 1$ choices.

The same rectangle starts at A , B , C , or D . Each rectangle is counted **exactly four times**.

$$\frac{n^2(n-1)^2}{4} \text{ rectangles}$$

Exercise: Squares on a grid



Shown: 5×5 points

On the same $n \times n$ grid of equally spaced **points** ($n \geq 2$):

How many different squares are there?

Vertices must be grid points. Only horizontal and vertical sides are allowed.

Left as an exercise.

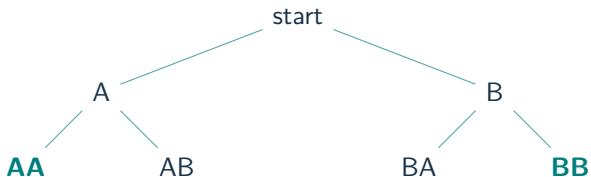
Tree diagrams

Teams A and B play a best-of-three series.

The first team to win two games wins.

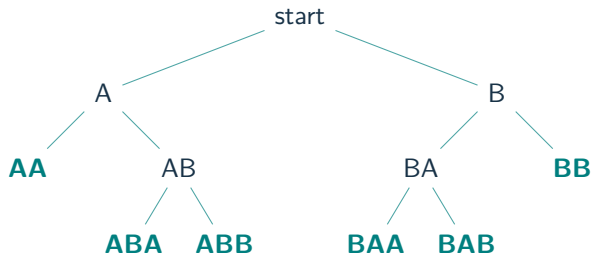
What complete game sequences are possible?

Tree diagrams



Which branches require a third game?

Tree diagrams



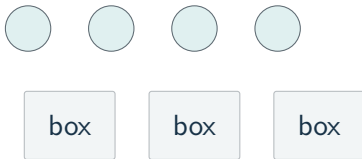
Each root-to-leaf path is one outcome. There are **six**.

SECTION 4

Guaranteed repetition with pigeonhole principles

Pigeonhole principle

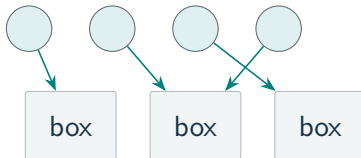
MUST UNDERSTAND More objects than boxes force a collision.



Can every box receive at most one object?

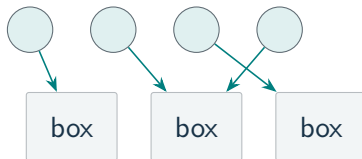
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Pigeonhole principle

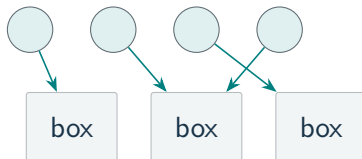
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If $k + 1$ objects are assigned to k boxes, some box receives at least two objects.

Pigeonhole principle

MUST UNDERSTAND More objects than boxes force a collision.



If $k + 1$ objects are assigned to k boxes, some box receives at least two objects.

If every box held at most one object, the boxes could hold at most k objects.

Objects and boxes

27 English words

Objects: the 27 words.

Boxes: 26 initial letters.

Objects and boxes

27 English words

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Two words share an initial letter.

Objects and boxes

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Two words share an initial letter.

A function $U \rightarrow V$

Objects: elements of U .

Boxes: elements of V .

If $|U| > |V|$, the function is not injective:

$$\exists x, y \in U, \quad x \neq y, \quad f(x) = f(y).$$

Two distinct inputs **collide**.

Generalized pigeonhole principle

Assign n objects to k boxes, where $n \geq 0$ and $k \geq 1$.

Some box contains at least $\left\lceil \frac{n}{k} \right\rceil$ objects.

Generalized pigeonhole principle

Assign n objects to k boxes, where $n \geq 0$ and $k \geq 1$.

Some box contains at least $\left\lceil \frac{n}{k} \right\rceil$ objects.

Example: $n = 10$, $k = 4$

$$\left\lceil \frac{10}{4} \right\rceil = 3.$$

If every box held at most 2, four boxes could hold at most 8.

capacity if every box held at most 2



Generalized pigeonhole principle

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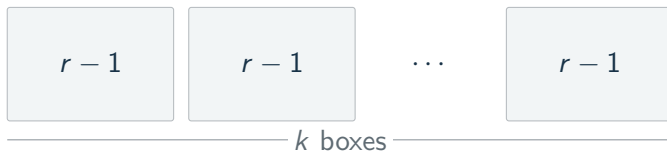
capacity if every box held at most 2



● cannot fit

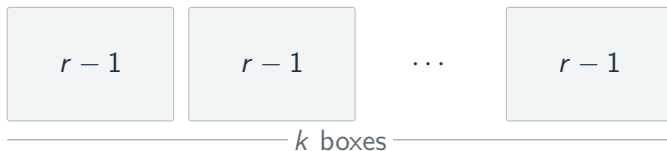
Forcing at least r objects

The largest arrangement that avoids r in every box has $k(r - 1)$ **objects**.



Forcing at least r objects

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Therefore,

$$k(r - 1) + 1$$

objects force at least r into one box.

The distribution need not be even

For $n = 10$ objects and $k = 4$ boxes,

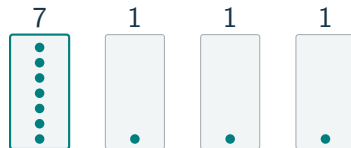
$$\left\lceil \frac{n}{k} \right\rceil = 3, \quad n \bmod k = 2.$$

Most even distribution



3, 3, 2, 2: two boxes reach 3.

A lopsided distribution



7, 1, 1, 1: only one box reaches 3.

The theorem guarantees **at least one** such box—not $n \bmod k$ such boxes.

Example 4: The most frequent combo

A customer orders one meal per day for 1000 days from 225 possible meal combos.

Soy sauce chicken + cabbage + milk tea is his **most frequently ordered combo**.

At least how many times did he order it?



Example 4: The most frequent combo

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Objects: 1000 meals

Boxes: 225 meal combos



Example 4: The most frequent combo



A customer orders one meal per day for 1000 days from 225 possible meal combos.

Soy sauce chicken + cabbage + milk tea is his **most frequently ordered combo**.

1000 meals are assigned to 225 combo boxes.

$$\left\lceil \frac{1000}{225} \right\rceil = 5.$$

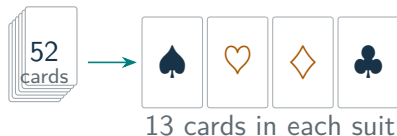
He ordered it **at least 5 times**.

Some box and a specified box

Draw cards from a standard 52-card deck **without replacement**.

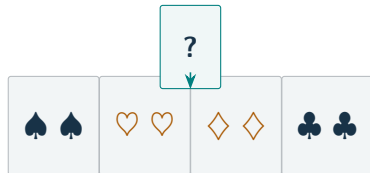
How many cards guarantee three of some suit?

How many guarantee three hearts?



Some box and a specified box

Three of some suit



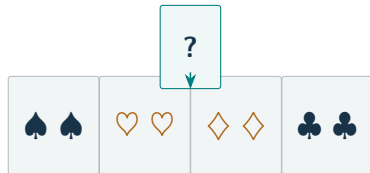
8 cards: two from each suit

The ninth card gives three of some suit.

$$4(3 - 1) + 1 = \boxed{9}.$$

Some box and a specified box

Three of some suit

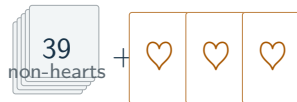


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Three hearts



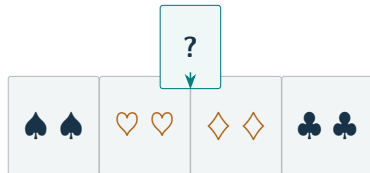
latest possible third heart

First draw all 39 non-hearts.

$$39 + 3 = \boxed{42}.$$

Some box and a specified box

Three of some suit



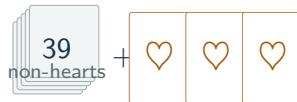
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“Some box” does not mean a box specified in advance.

Three hearts



latest possible third heart

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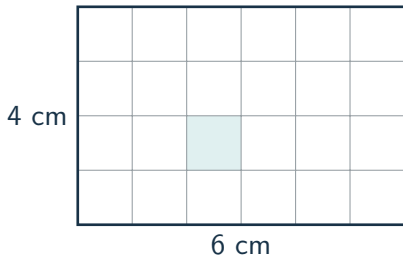
$$39 + 3 = \boxed{42}.$$

Geometric pigeonholes

Place 25 points anywhere inside a 6 cm by 4 cm rectangle.

Show that two points are less than 2 cm apart.

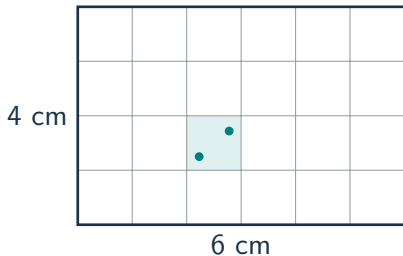
Geometric pigeonholes



24 boxes

Partition into unit squares;
assign boundary points
consistently.

Geometric pigeonholes

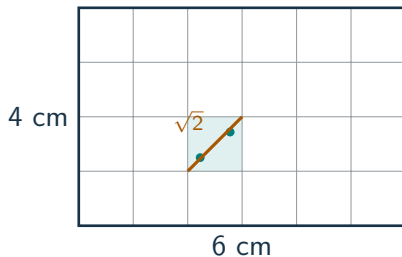


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consistently.

With 25 points, two share a
square.

Geometric pigeonholes



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With 25 points, two share a
square.

The diameter is

$$\sqrt{2} < 2 \text{ cm.}$$

SECTION 5

Problem solving

Exam problem: Registration codes

A registration system accepts codes of length 1, 2, or 3.

The first character is an uppercase letter.

Later characters may be uppercase letters or digits.

Twelve otherwise valid codes are reserved.

How many codes remain available?

Registration codes: Solution

Length	Count	Structure
1	26	one choice
2	$26 \cdot 36$	product
3	$26 \cdot 36^2$	product

Registration codes: Solution

Length	Count	Structure
1	26	one choice
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Add the disjoint lengths; subtract the reserved codes:

$$26 + 26 \cdot 36 + 26 \cdot 36^2 - 12 = \boxed{34,646}.$$

Exam problem: Queue threshold

A server distributes incoming requests among 12 queues.

How many requests guarantee at least 9 in one queue?

Exam problem: Queue threshold

Largest avoiding arrangement: at most 8 requests per queue.

$$12 \cdot 8 = 96.$$

Exam problem: Queue threshold

Largest avoiding arrangement: at most 8 requests per queue.

$$12 \cdot 8 = 96.$$

The next request forces a ninth request in some queue:

$$12(9 - 1) + 1 = \boxed{97}.$$

Practice: Choose the structure

1. Five-position decimal access codes; each position may be 0–9.
2. Travel by bus or ferry.
3. Six-character passwords with at least one digit.
4. Handshakes among six people.
5. A best-of-three playoff.
6. Students guaranteeing four with the same birth month.

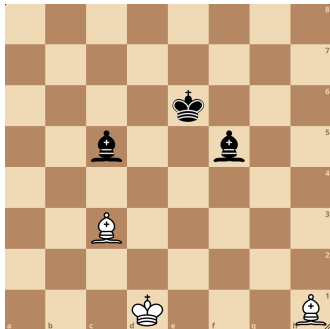
Practice: Choose the structure

Problem	Question	Structure
access code	exact	product
bus or ferry	exact	product + sum
password	exact	complement
handshakes	exact	division
playoff	exact	tree
birth months	guarantee	pigeonhole

Practice: Choose the structure

Problem	Structure	Result
access code	product	10^5
bus or ferry	product + sum	18
password	complement	$36^6 - 26^6$
handshakes	division	$6 \cdot 5/2 = 15$
playoff	tree	6
birth months	pigeonhole	$12(4 - 1) + 1 = 37$

Challenge: Two kings and four bishops



32 light squares and 32 dark squares.

On an 8×8 chessboard, each player has:

- one king;
- one bishop on a light square;
- one bishop on a dark square.

White's and Black's pieces are distinct. All six pieces occupy different squares.

Ignore checks, king adjacency, and other chess rules for now.

How many different placements are there?

Roadmap: Counting six-piece placements

1. Define one outcome.

A square for each of six distinct pieces; no two share a square. Player and bishop square color distinguish the pieces: no division needed.

2. Identify the choices and choose an order.

Place the six pieces:

$$K_B, \quad K_W, \quad B_{B,L}, \quad B_{B,D}, \quad B_{W,L}, \quad B_{W,D}.$$

K : king; B : bishop. Subscripts: player (Black/White), then square color (Light/Dark).

3. Chain the choices with the product rule.

Count available squares at each step. Split into disjoint cases and add with the sum rule when needed.

Six-piece placements: Place the bishops first

Place White's piece before Black's within each row:

Pieces	Choices
Two light-squared bishops	$32 \cdot 31$
Two dark-squared bishops	$32 \cdot 31$

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Two kings on the remaining squares	$60 \cdot 59$

After placing the four bishops, exactly 60 squares remain.

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Two light-squared bishops	$32 \cdot 31$
Two dark-squared bishops	$32 \cdot 31$
Two kings on the remaining squares	$60 \cdot 59$

After placing the four bishops, exactly 60 squares remain.

$$T = (32 \cdot 31)^2 \cdot 60 \cdot 59 = \boxed{3,483,586,560}.$$

No division is needed: player and square color identify each piece.

Choosing the order: What if kings go first?

The kings' square colors determine how many squares remain for each bishop pair.

White king	Black king	King placements	Light bishops	Dark bishops
light	light	$32 \cdot 31$	$30 \cdot 29$	$32 \cdot 31$
dark	dark	$32 \cdot 31$	$32 \cdot 31$	$30 \cdot 29$
light	dark	$32 \cdot 32$	$31 \cdot 30$	$31 \cdot 30$
dark	light	$32 \cdot 32$	$31 \cdot 30$	$31 \cdot 30$

Multiply across each row, then add the four disjoint cases.

Choosing the order: What if kings go first?

The kings' square colors determine how many squares remain for each bishop pair.

White king	Black king	King placements	Light bishops	Dark bishops
light	light	$32 \cdot 31$	$30 \cdot 29$	$32 \cdot 31$
dark	dark	$32 \cdot 31$	$32 \cdot 31$	$30 \cdot 29$
light	dark	$32 \cdot 32$	$31 \cdot 30$	$31 \cdot 30$
dark	light	$32 \cdot 32$	$31 \cdot 30$	$31 \cdot 30$

Multiply across each row, then add the four disjoint cases.

$$\begin{aligned} T &= 2(32 \cdot 31)^2(30 \cdot 29) + 2 \cdot 32^2(31 \cdot 30)^2 \\ &= \boxed{3,483,586,560}. \end{aligned}$$

Bishops first: the kings always have $60 \cdot 59$ choices; no case split.

Follow-up: Kings may not share an edge

New restriction: the two kings may not occupy squares sharing an edge. Diagonal neighbors are allowed in this exercise. Keep all other assumptions.



one light + one dark

How many placements remain?

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How many placements remain?

Count the forbidden placements first: place the edge-adjacent kings, then the bishops.

Counting the forbidden placements

A pair of edge-adjacent squares forms a 1×2 domino.

Horizontal domino locations $8 \cdot 7 = 56$

Vertical domino locations $8 \cdot 7 = 56$

Assign White's and Black's kings 2 orders per location

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Each pair uses one light and one dark square, leaving 31 of each.

$$\text{Bishop placements} = (31 \cdot 30)^2.$$

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$$F = (56 + 56) \cdot 2 \cdot (31 \cdot 30)^2 = 193,737,600.$$

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$$F = (56 + 56) \cdot 2 \cdot (31 \cdot 30)^2 = 193,737,600.$$

$$T - F = 3,483,586,560 - 193,737,600 = \boxed{3,289,848,960}.$$

Takeaways

Exact count

Describe one complete outcome.

- successive choices $\rightarrow$ product;
- disjoint cases $\rightarrow$ sum;
- reorganize with complement, uniform division, or a tree.

Guaranteed repetition

Name the objects and boxes.

- find the largest arrangement that avoids the target;
- add one to force the target.